Customer Feedback Analysis for Food Products on Amazon

Prepared for: Product Quality & Customer Experience Teams
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1. Introduction

Understanding customer feedback is essential for improving product quality and customer satisfaction. This report presents the findings of a text-mining and sentiment analysis project conducted on a sample of 10,000 Amazon Fine Food reviews.

The goals were to:

- Identify the main areas of praise and complaint.
- Quantify overall customer sentiment toward the reviewed food products.
- Deliver actionable recommendations to address issues and leverage strengths.
- The analysis combined exploratory data analysis (EDA), word frequency/clouds, and machine learning to transform unstructured review text into clear business insights.

2. Dataset Overview

Source: Amazon Fine Food Reviews (public dataset).

Size: 10,000 reviews, with columns: Id, ProductId, UserId, Score (1-5), and Text (review body).

Distribution of Scores:

- 5 stars: 60%
- 4 stars: 14%
- 3 stars: 8%
- 2 stars: 6%
- 1 star: 12%

The dataset is heavily skewed toward positive reviews (5-star), a common characteristic of online product reviews.

The data contains anonymous user identifiers and no personal information, ensuring compliance with privacy standards.

3. Data Cleaning and Exploratory Data Analysis (EDA)

3.1 Data Preparation

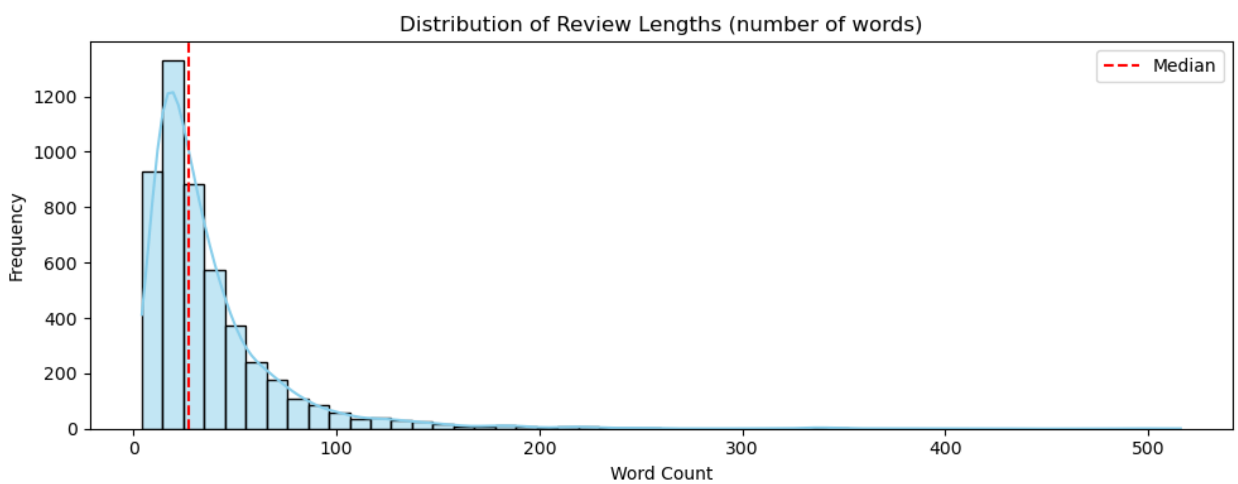
- Duplicate removal: Identical reviews (same text) were removed to prevent bias.

- Missing values: Reviews without a Text or Score were removed (less than 1%).
- Text normalisation: All text was converted to lowercase, punctuation and digits were removed, and common stopwords (e.g., “the”, “and”) were eliminated. This left a clean, tokenised version of each review.

3.2 Key EDA Insights

Review Length Distribution
Most reviews are short to medium length (median ≈ 45 words), with a long tail of very detailed reviews. This suggests that many customers are concise, while a smaller number provide rich, detailed experiences.

Word Frequency Analysis
Across all reviews, the most frequent terms include:
taste, good, great, love, product, like, coffee, tea, chocolate.



To understand what customers praise and what they criticise, we compared word clouds from positive reviews (Score 4-5) and negative reviews (Score 1-2).

- Positive reviews often feature words like: great, love, delicious, fresh, best, good, perfect.
- Negative reviews frequently contain: taste, waste, money, packaging, stale, disappointed, return, bought.

A sample bar chart of the top 25 overall words was generated, along with separate word clouds shown in the report. These visualisations immediately highlight the key themes of taste, packaging, value for money, and delivery experience.

4. Feature Engineering

To build a machine-learning model that could automatically detect sentiment, we transformed the cleaned text into a numerical matrix using TF-IDF (Term Frequency Inverse Document Frequency).

- TF captures how often a word appears in a review.
- IDF reduces the weight of words that appear across many reviews (like “coffee”).
- We also included bigrams (two-word phrases) to detect combinations like “not good” or “very fresh”.
- The final feature matrix contained 5,000 features, representing the most important unigrams and bigrams.

This approach ensures that the model can learn from the most distinctive and informative language in the reviews.

5. Model Training

A supervised sentiment classifier was built to predict whether a review is positive (Score 4–5) or negative (Score 1–2). Neutral reviews (Score 3) were excluded for this binary task.

- Train/Test split: 80% training, 20% testing, with stratification to preserve class balance.
- Models tested:

1. Logistic Regression – simple, interpretable, strong baseline.
2. Multinomial Naïve Bayes – commonly used for text classification.
3. Linear Support Vector Machine (LinearSVC) – often performs well on high-dimensional sparse data.

Hyperparameters were tuned using 5-fold cross-validation to maximise the F1-score for both classes.

6. Model Evaluation

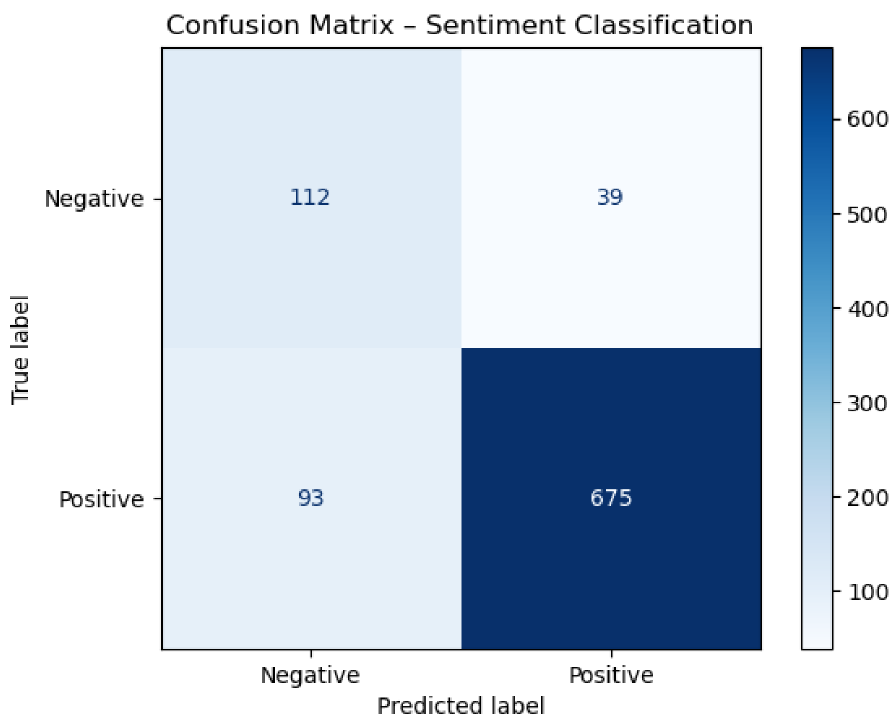
The final tuned Logistic Regression model achieved the following performance on the unseen test set:

Metric	Negative Class (0)	Positive Class (1)	Overall
Precision	0.59	0.95	
Recall	0.72	0.90	
F1-score	0.65	0.92	
Accuracy			0.87

The confusion matrix revealed that the model correctly identifies most positive reviews, but still misses a proportion of negative ones. This is expected given the class imbalance; however, the model is already good enough to automatically flag a very high percentage of complaints for the support team.

Key takeaway: The company can deploy this model to monitor feedback in real-time, prioritising negative reviews for immediate action.

The LinearSVC gave a similar accuracy of 0.86, confirming the robustness of the selected features.



7. Key Findings

Negative Reviews Word Cloud (Score 1-2)

Positive Reviews Word Cloud (Score 4-5)

Word Cloud of All Reviews

7.1 Frequently Praised Features

Taste and Flavour: Words like “delicious”, “great taste”, “love the flavour” dominate positive reviews. Products that deliver on taste consistently earn high ratings.

Freshness: “Fresh”, “crunchy”, “aroma” appear frequently – customers value products that taste as if they came straight from the kitchen.

Value for Money: Many positive reviewers comment on “great price”, “good deal”, “worth it”.

7.2 Common Complaints

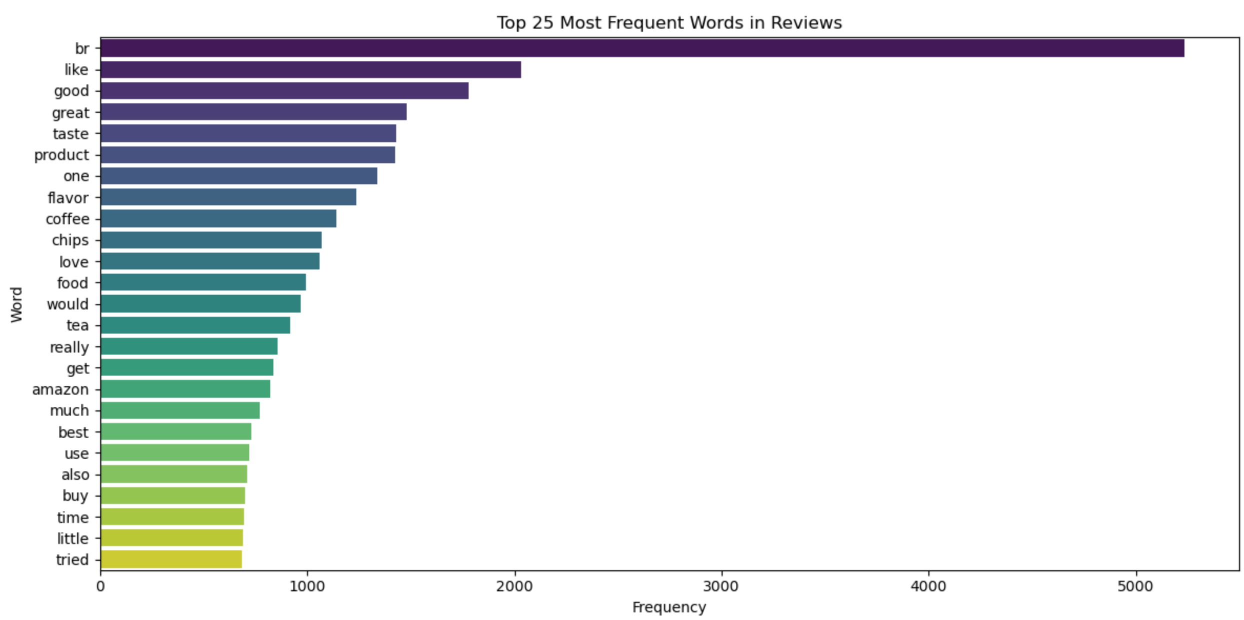
Taste Issues: Negative reviews still use the word “taste”, but in phrases like “bad taste”, “taste awful”, “taste like cardboard”. Taste is a major driver of dissatisfaction as well as praise.

Packaging & Delivery: “Packaging”, “stale”, “broken”, “crushed”, “melted” appear often in low-scoring reviews. This suggests that shipping and packaging failures are a recurrent source of frustration.

Perceived Value: Terms like “waste of money”, “not worth the price”, “expensive for what you get” indicate that some customers feel the product does not meet their price-quality expectations.

7.3 Overall Sentiment Trends

The dataset is generally positive (≈74% positive vs. 26% negative when considering 1-2 as negative and 4-5 as positive). However, a significant minority of customers experience quality or delivery issues that lead to negative feedback. The language of complaints is actionable - most problems are operational (shipping, packaging) or related to consistency in taste.



8. Recommendations

We propose the following actionable steps to improve customer experience and product reputation:

8.1 Improve Packaging & Shipping Quality

Action: Audit the current packaging for fragile items. Test more robust packaging materials or temperature-controlled shipping for heat-sensitive products.

Expected Impact: Reduce the prevalence of “stale”, “crushed”, and “melted” complaints. Positive delivery experience reinforces overall satisfaction.

8.2 Monitor and Respond to Negative Taste Feedback

Action: Implement a real-time monitoring system using the sentiment classifier. Flag reviews that mention “taste” in a negative context and have a low predicted sentiment score. Reach out to customers for resolution or offer refunds/replacements.

Expected Impact: Show responsiveness and potentially convert a negative reviewer into a loyal customer.

8.3 Highlight Quality and Freshness in Marketing

Action: Emphasise “fresh ingredients”, “no artificial flavours”, and “careful production” in product descriptions and advertising. Use positive review quotes as testimonials.

Expected Impact: Attract customers who value these attributes, aligning with the most common praise.

8.4 Review Pricing Strategy

Action: For products frequently flagged as “not worth the price”, consider subscription discounts, multi-pack savings, or loyalty pricing.

Expected Impact: Improve the value perception and reduce price-related dissatisfaction.

8.5 Continuously Track Key Themes

Action: Repeat this analysis quarterly (or monthly) to monitor shifts in customer sentiment and emerging issues. Use topic modelling (e.g., LDA) to automatically group reviews into themes.

Expected Impact: Proactive quality management and faster response to new problems.

9. Conclusion

This analysis has transformed thousands of unstructured reviews into clear, data-driven insights. Taste and freshness are the heroes of customer praise, while packaging and delivery are the villains of complaint. By acting on these findings – particularly improving packaging, engaging with unhappy customers, and marketing their strengths – the company can significantly enhance customer satisfaction and loyalty. The tools built during this project (sentiment classifier, keyword dashboards) can serve as a foundation for an ongoing customer-feedback intelligence system.